

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 15: Delay Locked Loops

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Outline

- What is delay locked loop (DLL) and its applications
- DLL topologies
- DLL building blocks
- DLL model and transfer function
- Phase noise in DLL
- Frequency multiplying DLL (MDLL)

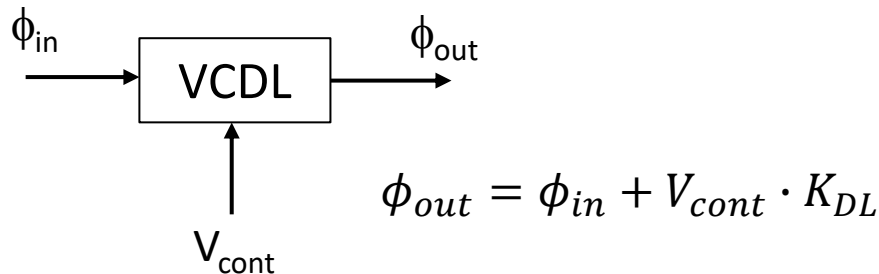
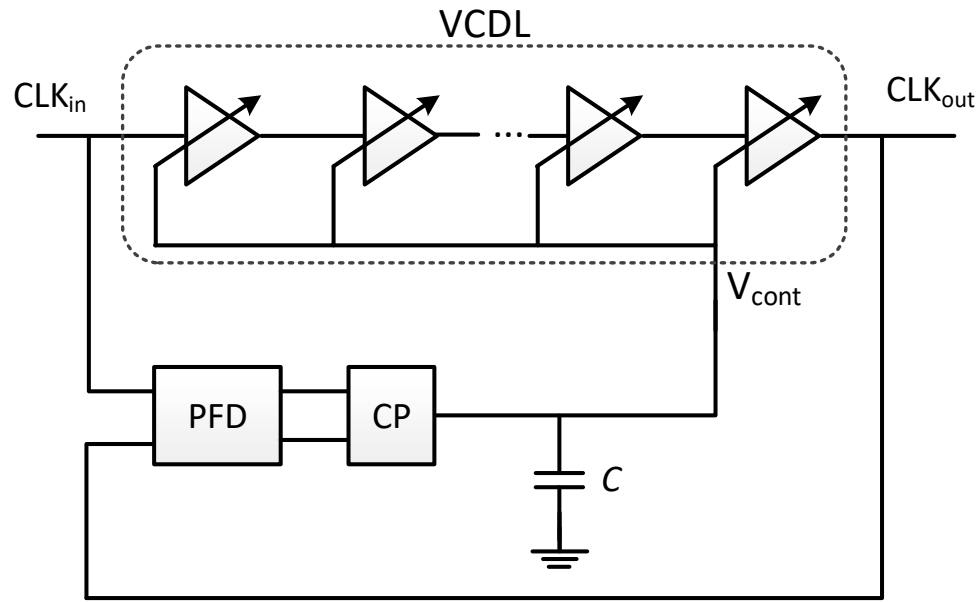


Delay Locked Loop Applications

- What is a delay locked loop (DLL)?
 - A DLL is similar to a PLL with the main difference that a VCO or DCO is replaced by a delay line.
- Why DLL
 - Generate an output signal with phase aligned with the input signal
 - Clock and data recovery (CDR)
 - Generate multi-phase signal
 - Frequency multiplication



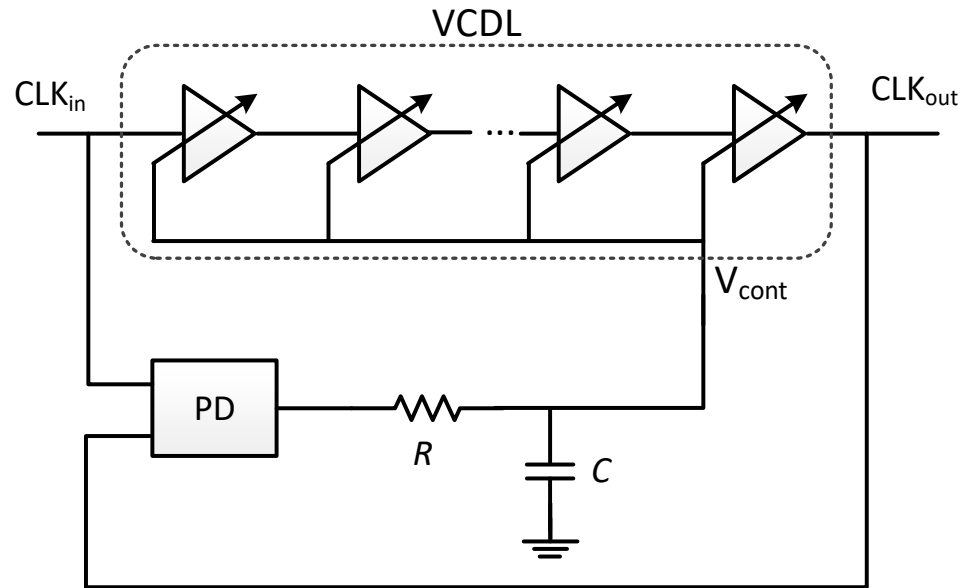
Delay Locked Loop Topology – PFD/CP Based DLL



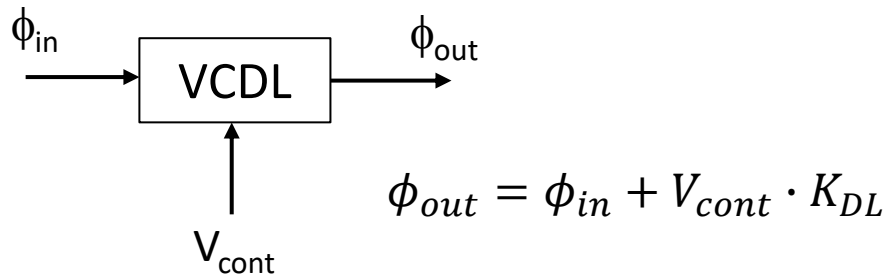
- Voltage controlled delay line (VCDL)
 - Generates delay ΔT or $\Delta \phi$ from CLK_{in} to CLK_{out}
 - $\Delta \phi = V_{cont} \cdot K_{DL}$
 - $\phi_{out} = \phi_{in} + V_{cont} \cdot K_{DL}$
- If there is any phase difference between CLK_{in} and CLK_{out} , PFD and charge pump will generate current pulses proportional to the phase difference and changes V_{cont} , which changes the delay and ϕ_{out} .
- Due to the negative feedback of the loop and the integrator from the capacitor, in steady state
 - $\phi_{out} = \phi_{in} + 2\pi$
 - Total delay is one clock cycle, $V_{cont} \cdot K_{DL} = 2\pi$



Delay Locked Loop Topology – PD Based DLL

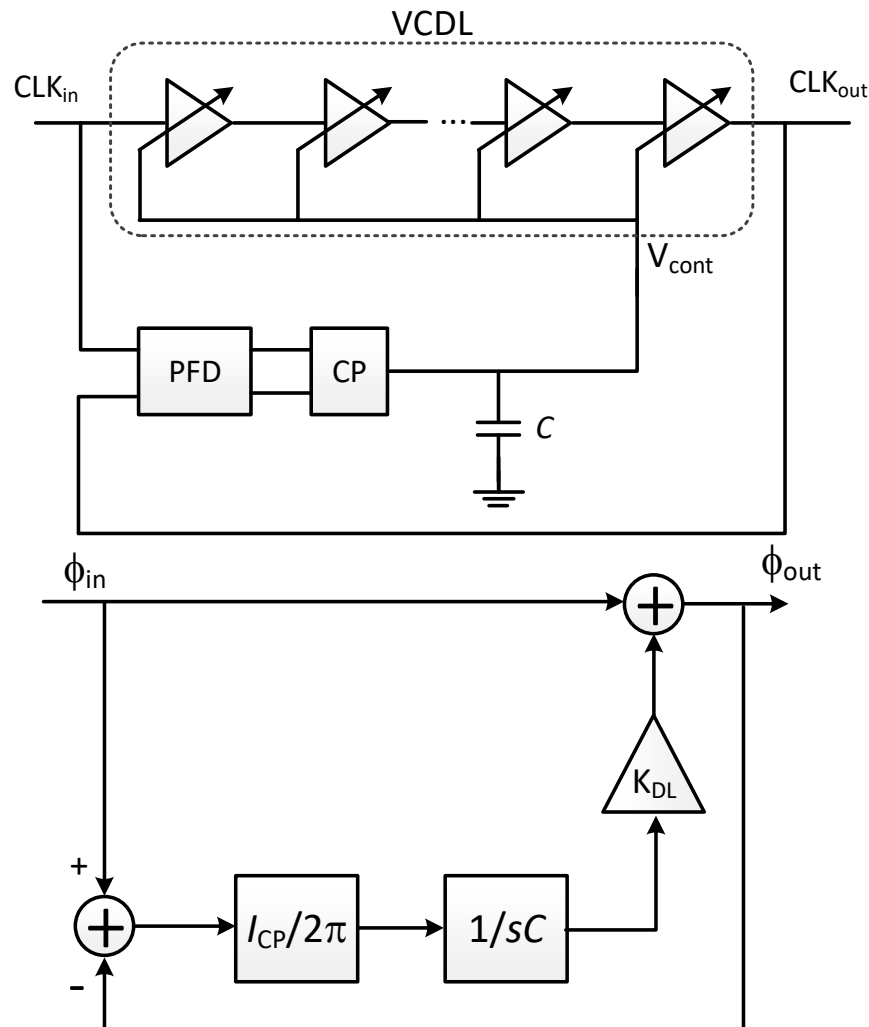


- Use a phase detector (PD) and RC filter instead of PFD and charge pump.
- In the steady state, there is a finite phase difference between CLK_{in} and CLK_{out} that is inversely proportional to PD gain K_{PD} and delay line gain K_{DL} .



Delay Locked Loop S-Domain Model

PFD/CP based DLL



$$\phi_{in} + (\phi_{in} - \phi_{out}) \frac{I_{CP}}{2\pi} \cdot \frac{K_{DL}}{sC} = \phi_{out}$$

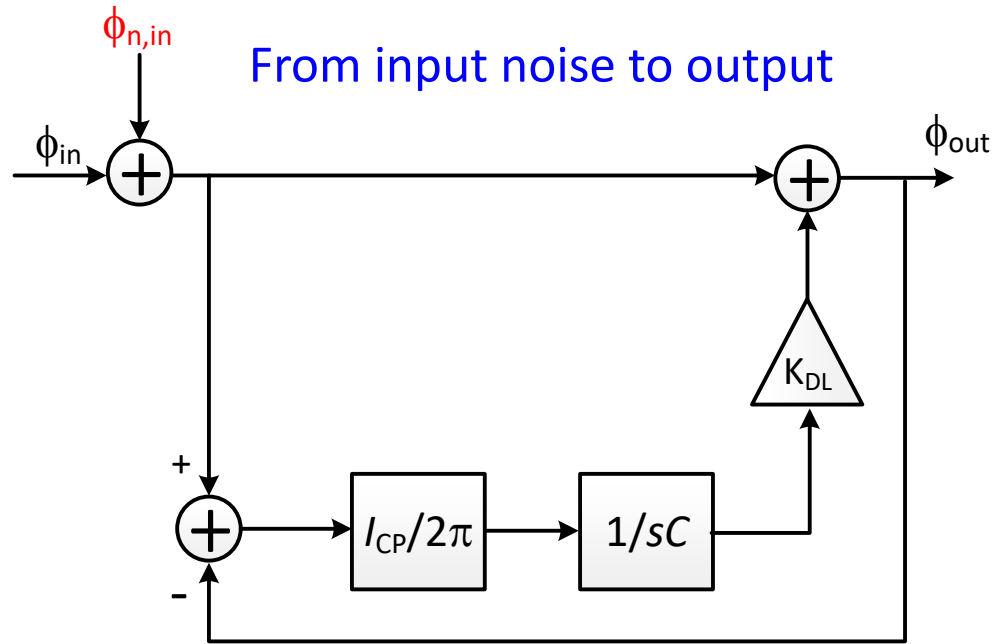
$$\phi_{in} - \phi_{out} = -(\phi_{in} - \phi_{out}) \frac{I_{CP}}{2\pi} \cdot \frac{1}{sC}$$

$$\Rightarrow \phi_{in} = \phi_{out}$$

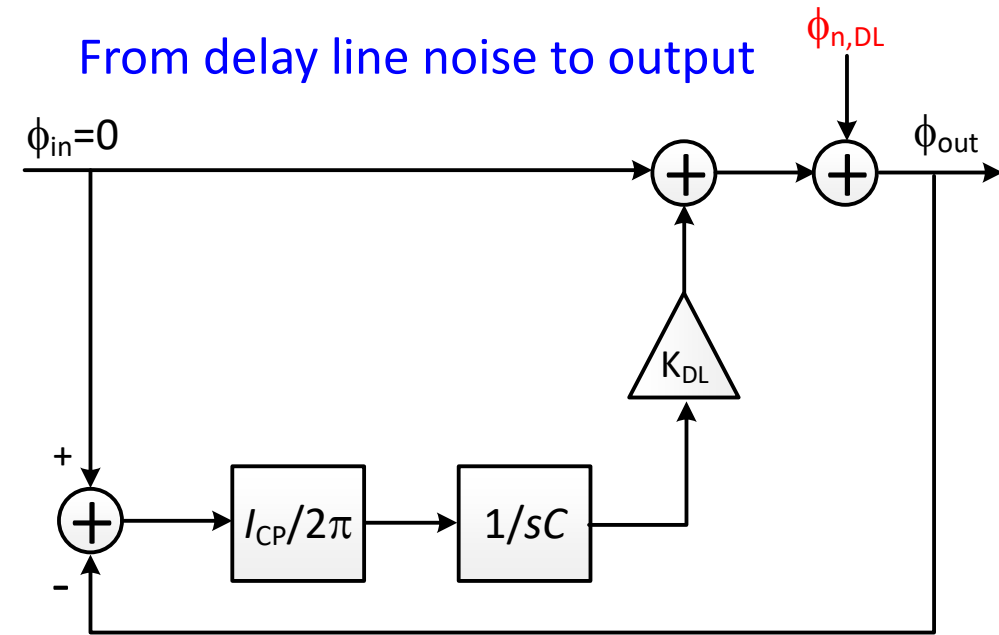
- All pass transfer function
- At low frequency, the loop gain is high, ϕ_{out} tracks ϕ_{in} and closed-loop transfer function has a unity magnitude.
- At high frequency, the loop gain becomes small, V_{cont} does not respond to input phase change, ϕ_{in} simply propagates to ϕ_{out} .



Phase Noise Transfer Functions



- Phase noise TF from input $\phi_{n,in}$ to output follows the same TF as signal ϕ_{in} .
- $\phi_{out} = \phi_{n,in}$



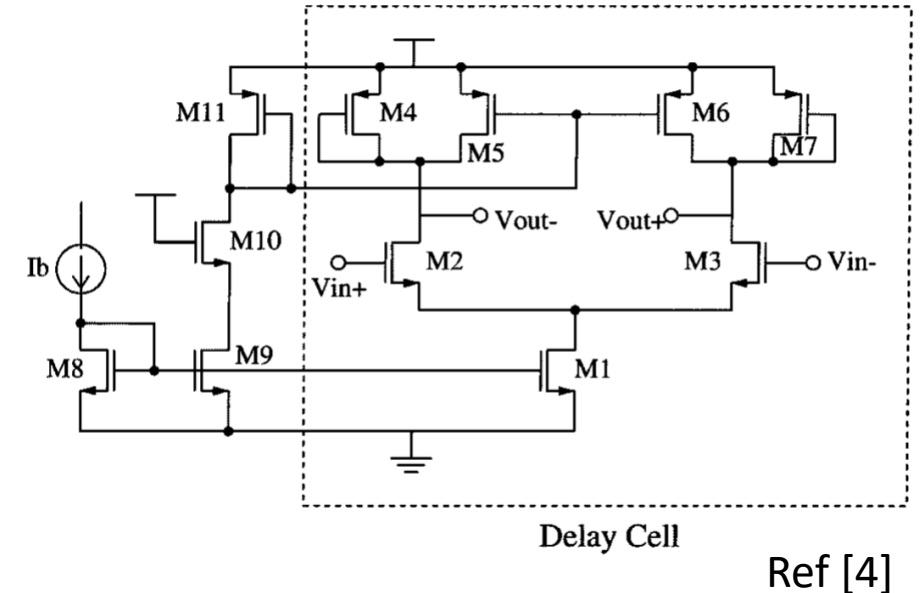
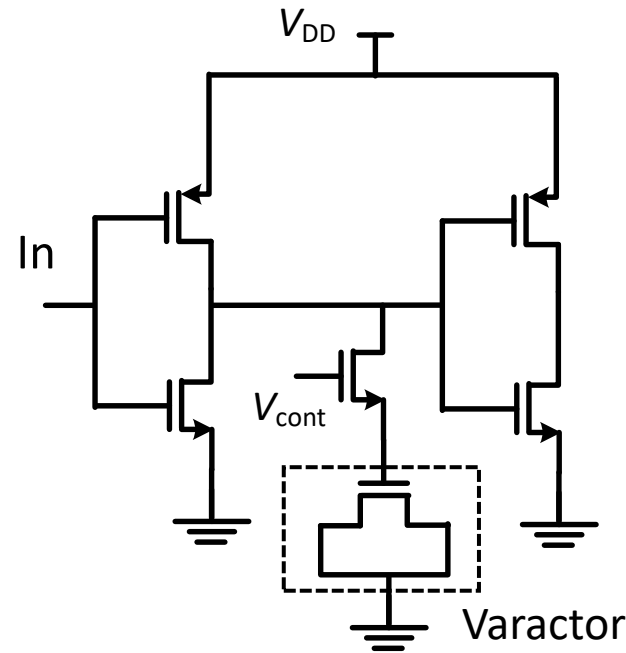
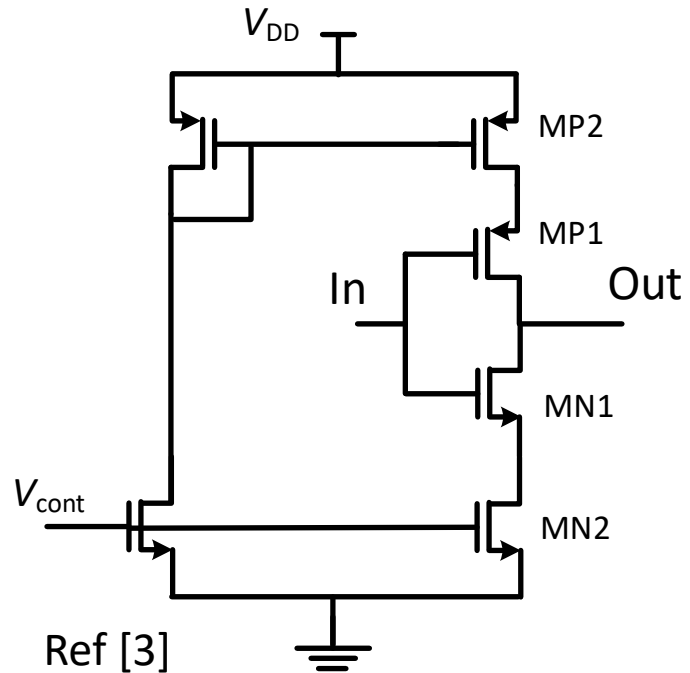
$$-\phi_{out} \left(\frac{I_{CP} \cdot K_{DL}}{2\pi \cdot sC} \right) + \phi_{n,DL} = \phi_{out}$$

$$\Rightarrow \frac{\phi_{out}}{\phi_{n,DL}} = \frac{1}{1 + \frac{I_{CP} \cdot K_{DL}}{2\pi \cdot sC}} = \frac{sC}{sC + \frac{I_{CP} \cdot K_{DL}}{2\pi}}$$

High pass transfer function



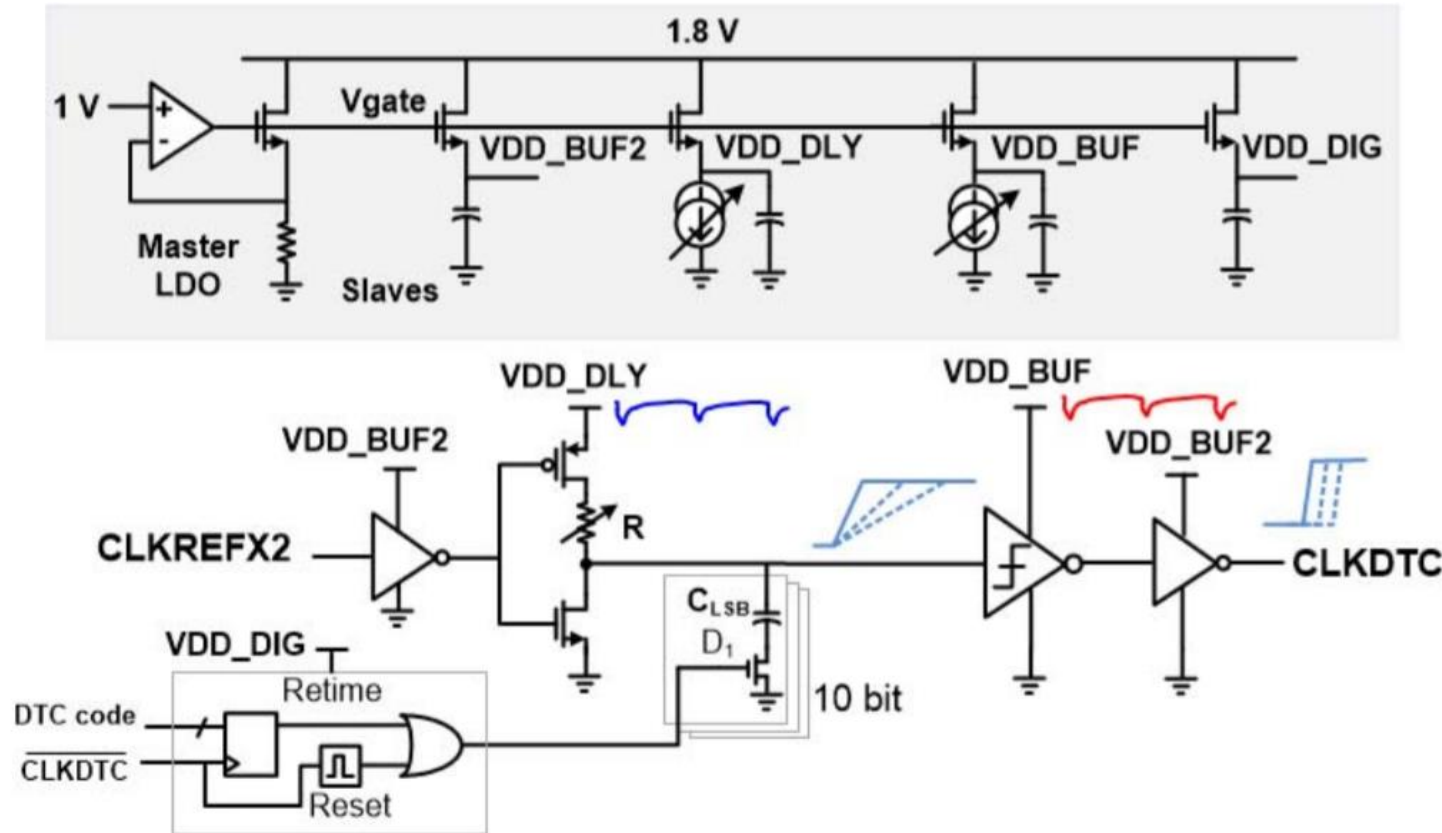
Voltage/Current Controlled Delay Cells



- Delay is determined by charge/discharge current and loading capacitance
- DLL delay cells and control are similar to those used in ring oscillators
 - Current starved inverter: delay is controlled by bias current through V_{cont}
 - Varactor inverter: delay is controlled by varactor capacitance
 - Differential delay cell: delay is controlled by bias current



Delay Cell based on RC Time Constant Control



$$V_C = V_{DD} \left(1 - e^{-\frac{t}{RC}} \right)$$

Time delay for V_C to reach inverter threshold voltage V_{th} :

$$t_d = RC \ln \left(\frac{V_{DD}}{V_{DD} - V_{th}} \right)$$

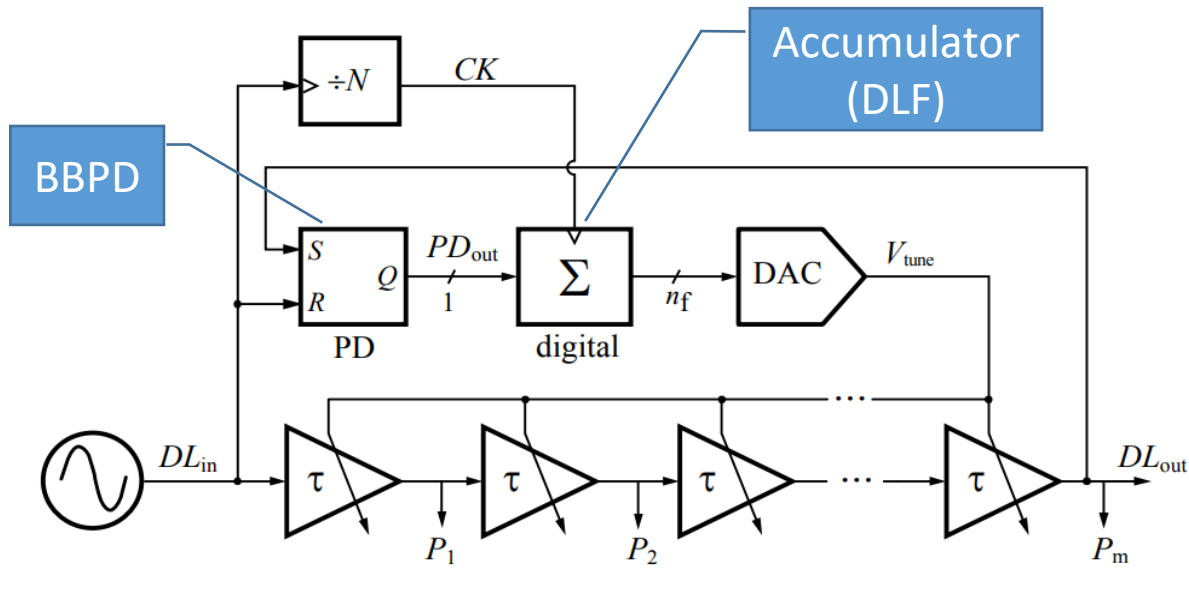
If $V_{th} = 0.5V_{DD}$,

$$t_d = RC \cdot \ln 2$$

- Voltage ramp slope determined by RC time constant
- RC time constant can be controlled controlled by changing the capacitance or resistance
- By changing either R or C, the delay can be changed.

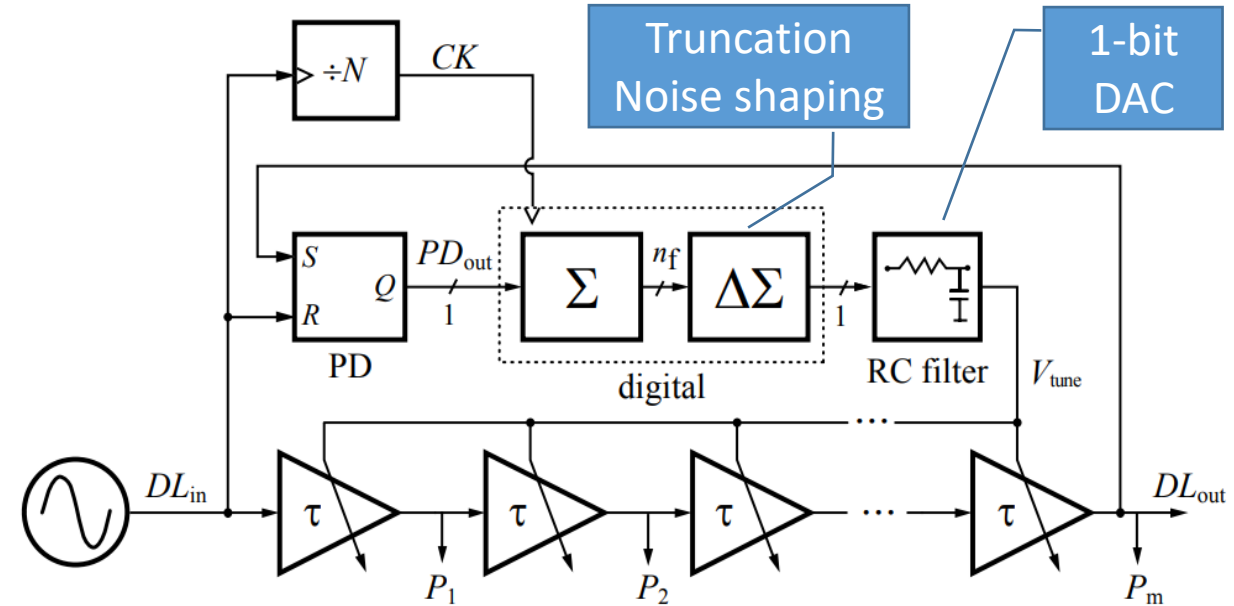


Digital Delay Locked Loop



- A SR latch as bang-bang phase detector (BBPD)
- An accumulator (integrator) as digital loop filter
- Clock DLF and DAC at a lower clock frequency (Div-N)

Ref [12]



- Simplify DAC to 1-bit and replace with a RC filter
- DSM for truncation to high-pass noise shaping and reduce quantization noise



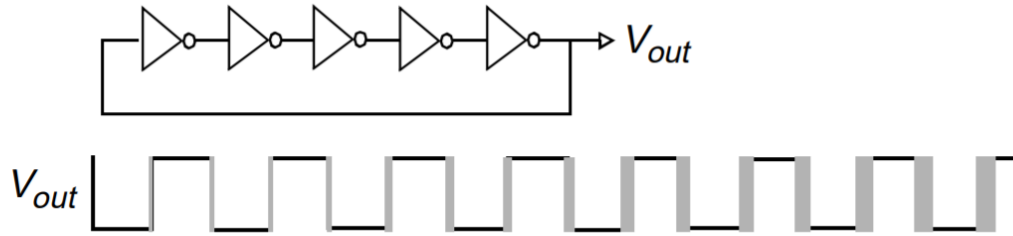
DLL Applications

- Multiphase generation
 - Time to digital conversion
 - Digital to time conversion
 - Generate quadrature phases in odd divide ratio dividers (e.g., Div-3)
- Frequency synthesis
 - Multiplying DLL (MDLL) for frequency multiplication
- Clock de-skew
 - To align the clock edges in distributed clock network

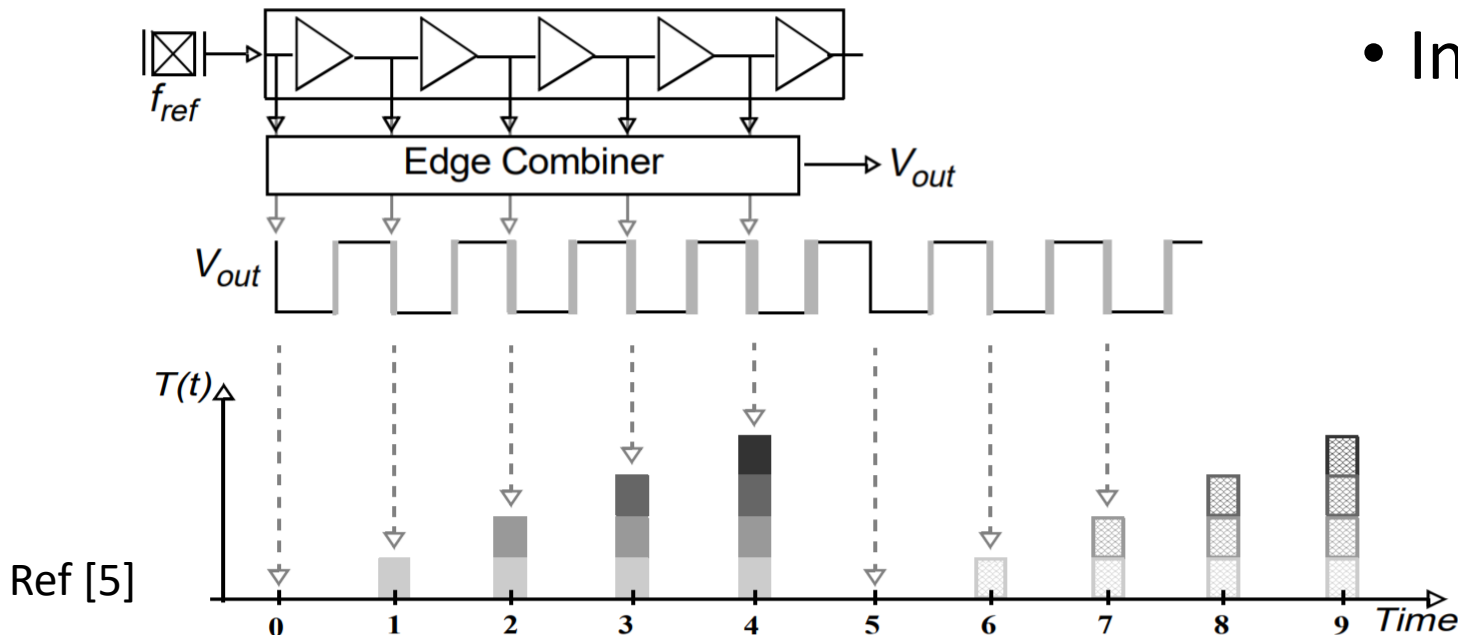


Timing Jitter in Ring Oscillator and Delay Line

Ring Oscillator



Delay Chain

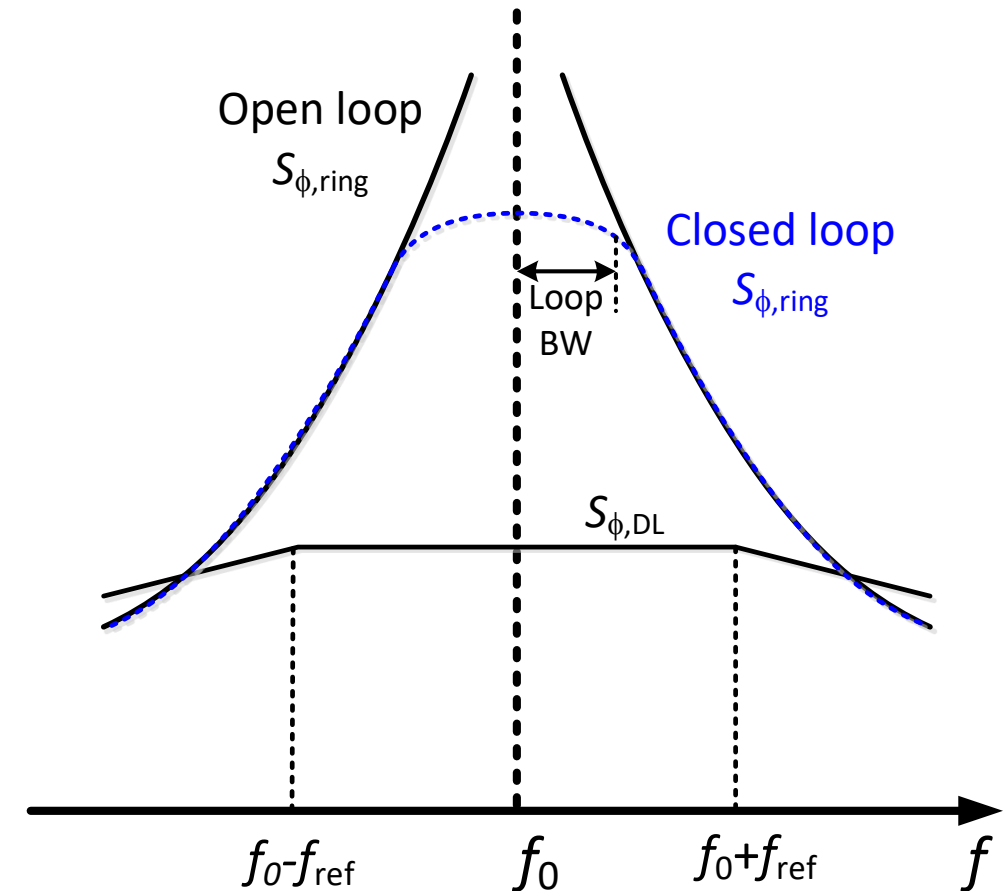


- In a ring oscillator
 - An edge continues to accumulate phase noise as it circulates around the loop.
 - Long term jitter performance is poor $\Rightarrow$ high close-in phase noise
- In a delay line
 - An edge accumulates up to N stages of phase noise. It is then synchronized with the low noise reference signal edge every reference cycle T_{ref} .
 - Low close-in phase noise up to offset frequency of F_{ref} .



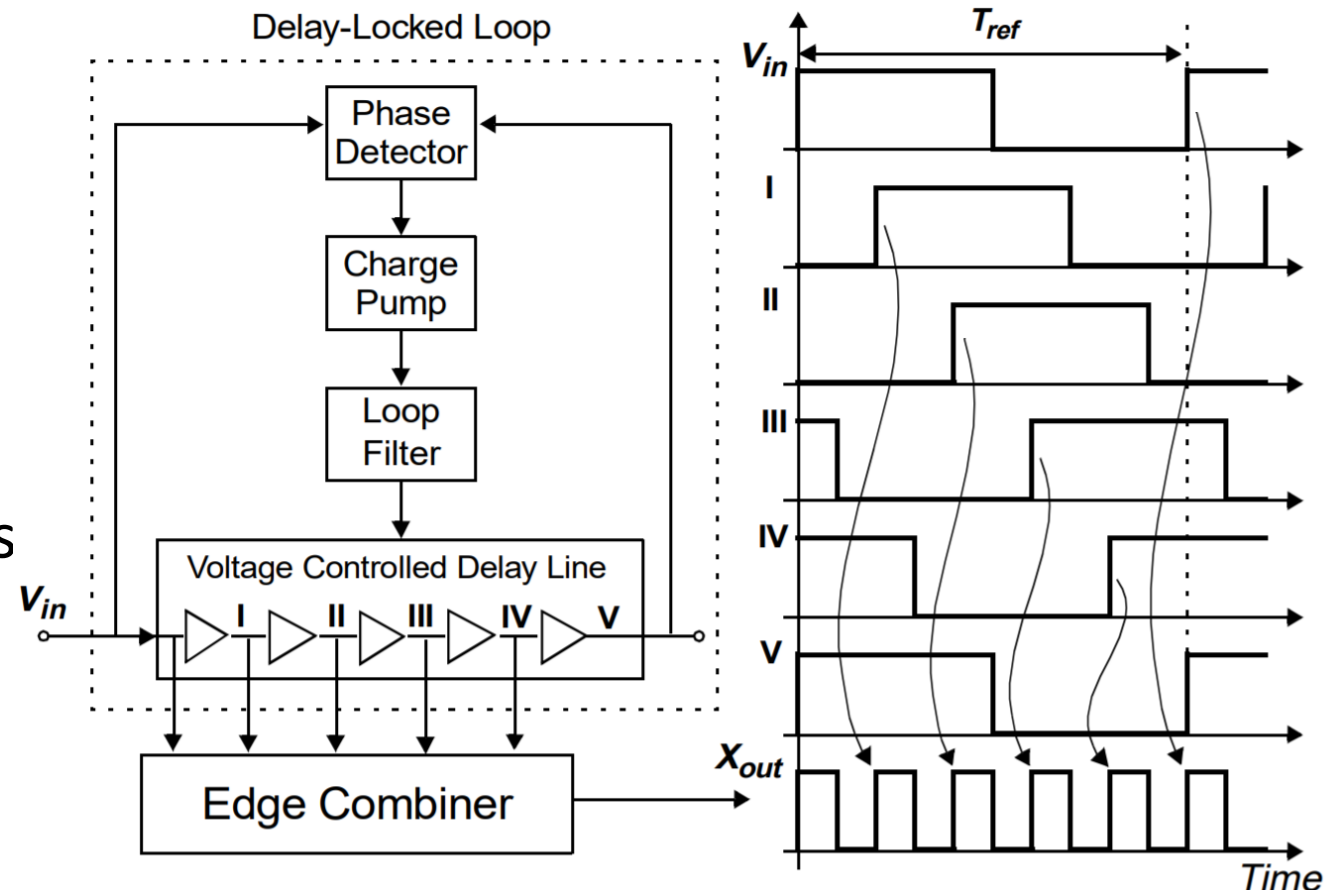
Phase Noise of Ring Oscillator and Delay Line

- Open loop $S_{\phi, \text{ring}}$: Ring oscillator has high closed-in phase noise.
- Closed-loop $S_{\phi, \text{ring}}$: PLL attenuates the phase noise within the loop bandwidth.
- Delay line $S_{\phi, \text{DL}}$
 - Within f_{ref} frequency offset, delay line phase noise is flat because for observation time longer than reference period, there is no inverter jitter accumulation beyond one delay chain jitter.
 - Phase noise reduces at higher offset frequency because for observation time shorter than reference period, the shorter the time (the higher the offset frequency), the less delay stages and less jitter accumulation.
- PLL loop BW is much smaller than f_{ref} , not sufficient in-band noise attenuation compared to DLL.



Multiplying DLL (MDLL) – Edge Combiner

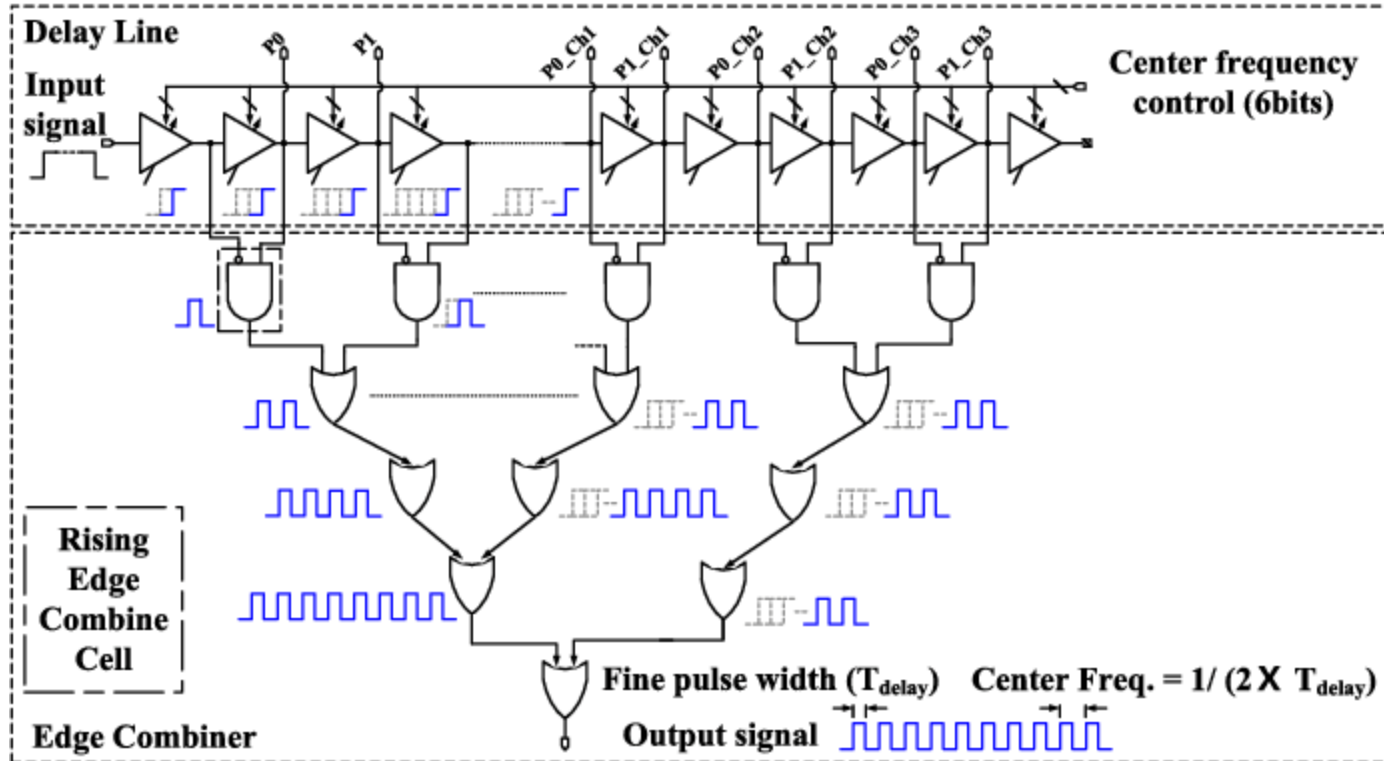
- Edge combiner MDLL
- DLL locks voltage controlled delay line (VCDL) to input clock frequency
- VCDL generates N -phase signals with unit time delay of T_{ref}/N .
- Edge combiner combines N signals at F_{in} to one signal at $N \cdot F_{\text{in}}$.
- Mismatch among delay stages causes spurs at subharmonics.
- Limited to integer multiplication ratio and difficult to program.



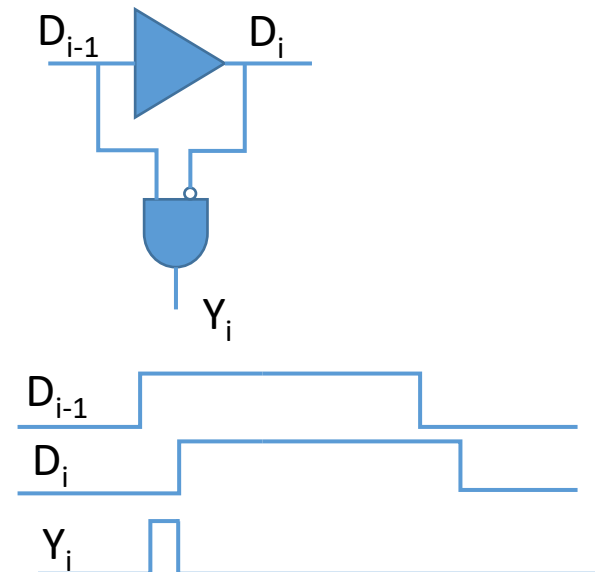
Ref [5]



Edge Combiner



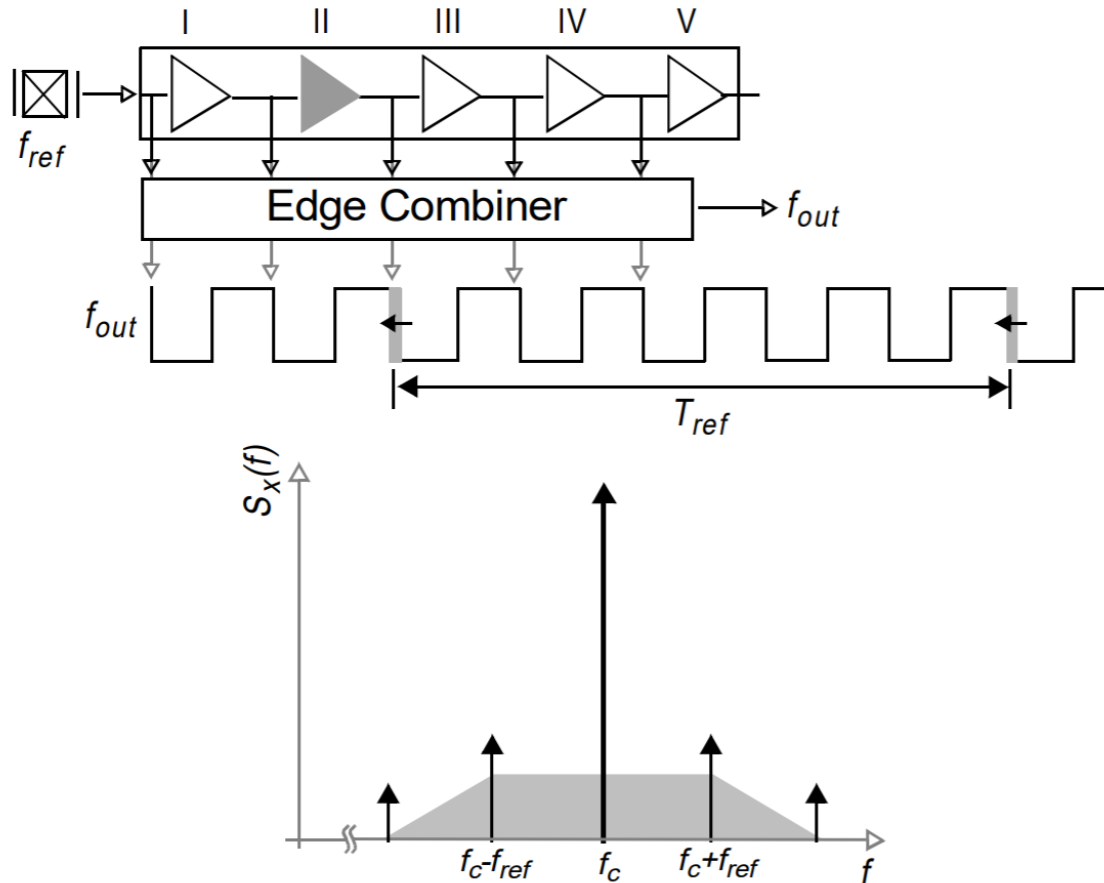
- Low frequency digital type of edge combiner
- Inverter, AND and OR gates based edge combiner



Ref [8]



Spurious Tones in Delay Locked Loop

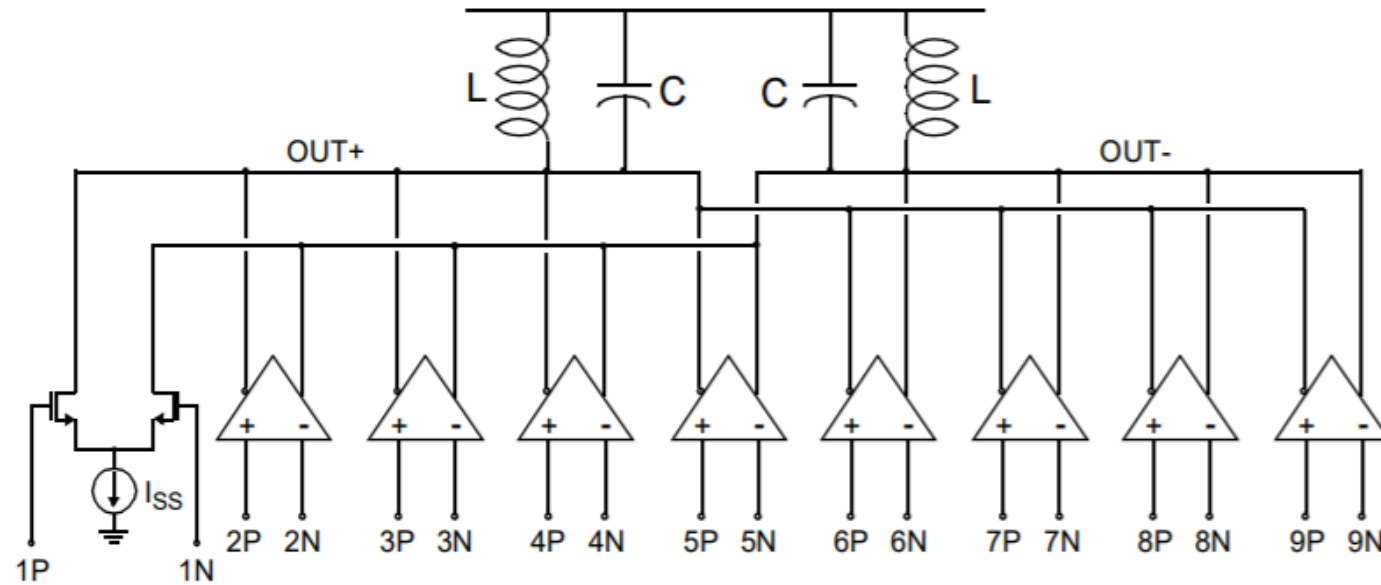


- Mismatch among delay cells generates periodic timing error at output signal with a period of T_{ref} .
- In frequency domain it shows up as spurs with an offset frequency at the harmonics of f_{ref} , $f_{out} \pm f_{ref}$, $f_{out} \pm 2f_{ref}$, etc.
- There are very strict spur level requirements in wireless communication transceivers in order not to desense the receiver or violate the transmit emission mask. Spur performance is one of the main concerns in edge combiner based MDLL frequency synthesis.

Ref [5]



Edge Combiner



Ref [9]

- High frequency RF type of edge combiner
 - N differential pairs, each input takes signal from the corresponding delay line tap
 - Differential pair outputs are combined and loaded by an LC tank tuned to f_{out} .
 - Mathematically equivalent to an N -tap FIR filter.



Edge Combiner

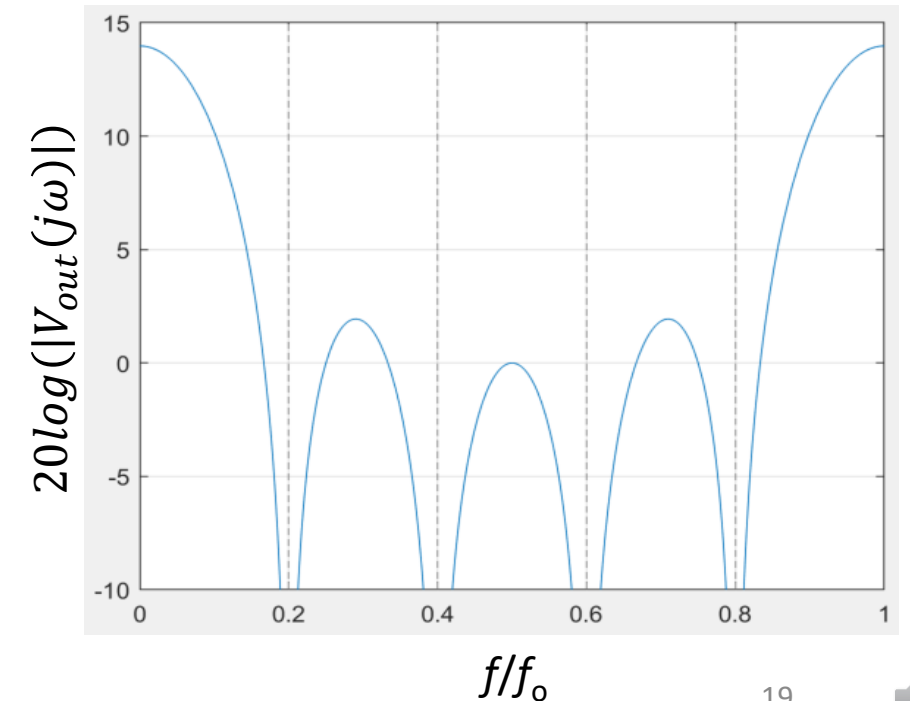
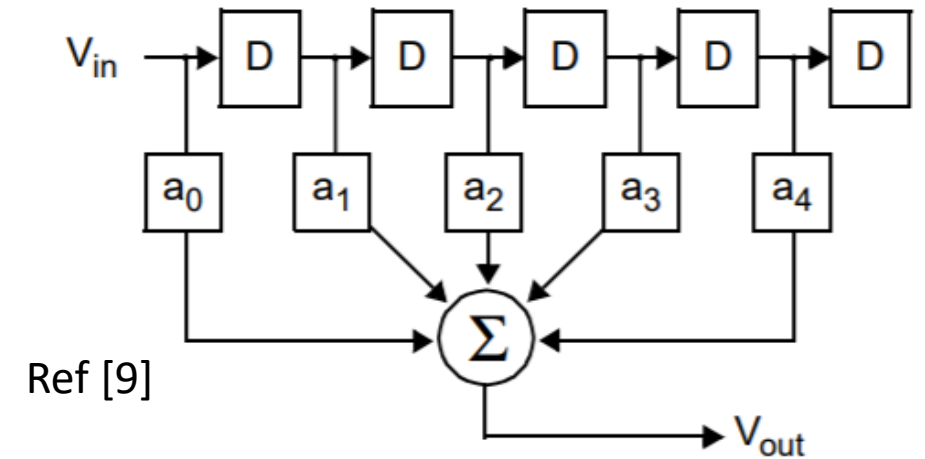
5-tap FIR filter example

$$\begin{aligned} V_{out}(j\omega) &= a_0 + a_1 e^{-j\omega/f_0} + a_2 e^{-j2\omega/f_0} + a_3 e^{-j3\omega/f_0} + a_4 e^{-j4\omega/f_0} \\ &= e^{-j2\omega/f_0} (e^{-j2\omega/f_0} + e^{-j\omega/f_0} + 1 + e^{j\omega/f_0} + e^{j2\omega/f_0}) \end{aligned}$$

Assuming all coefficients are equal.

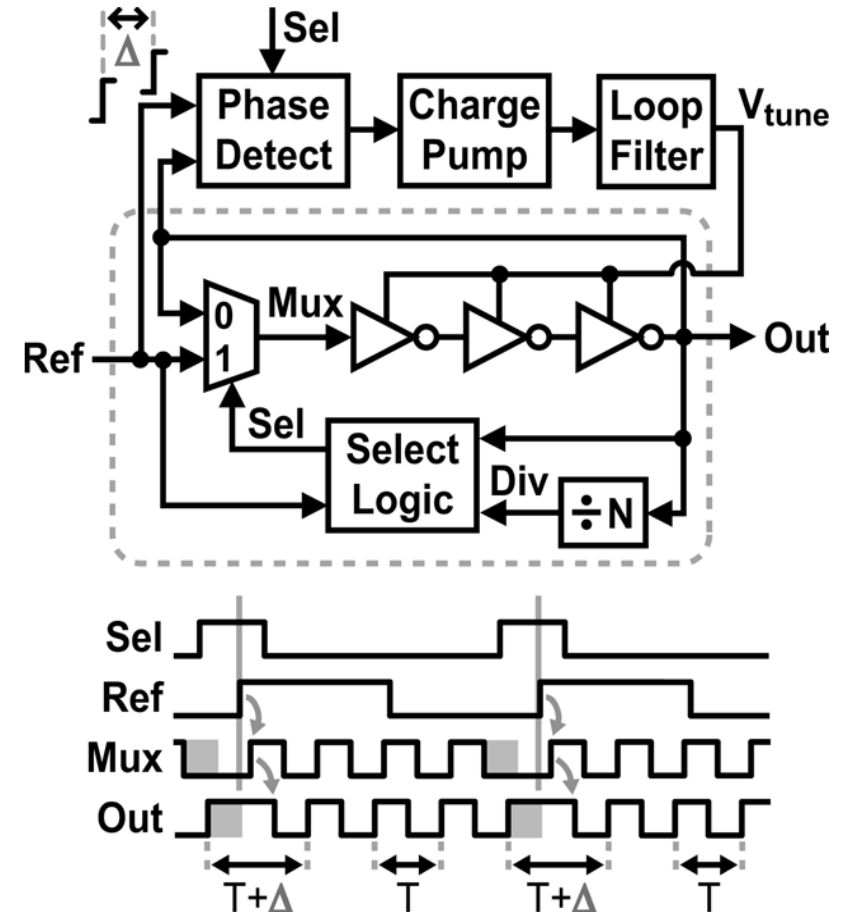
$$V_{out}(j\omega) = e^{-j2\omega/f_0} \left(1 + 2\cos\left(\frac{\omega}{f_0}\right) + 2\cos\left(\frac{2\omega}{f_0}\right) \right)$$

$|V_{out}(j\omega)|$ peaks at DC and harmonics of f_0 .
Subharmonics of $k/5f_0$ ($k=1, 2, 3, 4$) are suppressed.



Multiplying DLL (MDLL) – Edge Injected

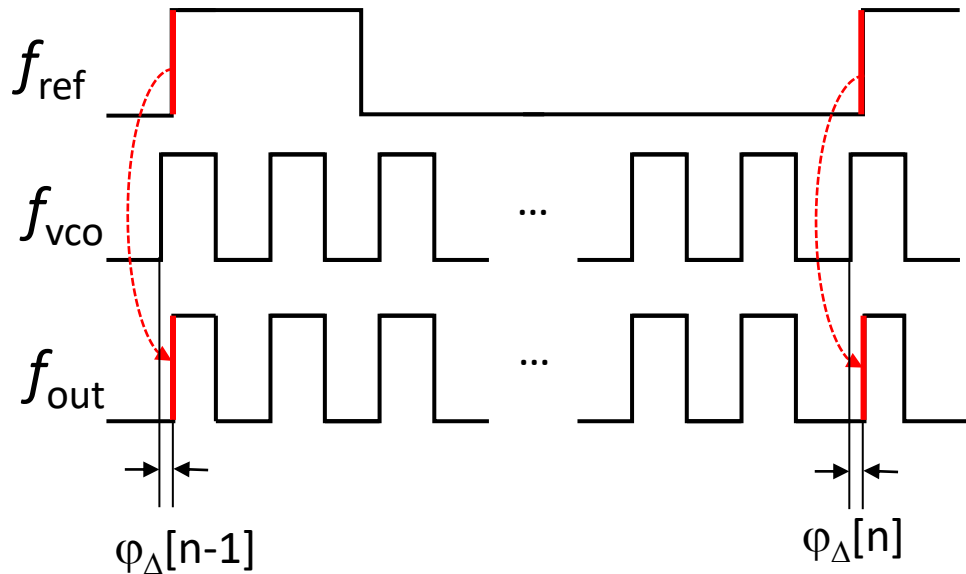
- Each rising edge of the input clock enters an inverting delay line via a multiplexer.
- After each edge passes, the multiplexer switches to select the output of the delay line such that a ring oscillator is formed.
- After passing $N-1$ edges, the multiplexer switches back to select the next rising edge of the reference signal as the N th edge.
- Meanwhile, the reference rising edge is compared with delay line output generating phase error that goes through charge pump and loop filter to generate the delay line control voltage to minimize the phase error.
- In the steady state, MDLL outputs N pulses in every one reference cycle, $f_{out} = N \cdot f_{ref}$.
- Multiplication ratio is limited to integer but it is easy to program by changing the divide ratio N .



Ref [6, 7, 10]



MDLL Phase Noise Transfer Functions



- In MDLL, ring oscillator noisy edges are periodically replaced with clean reference edges for every T_{ref} .
- Each replacement/realignment introduces a phase delta, which is accumulated to output phase.

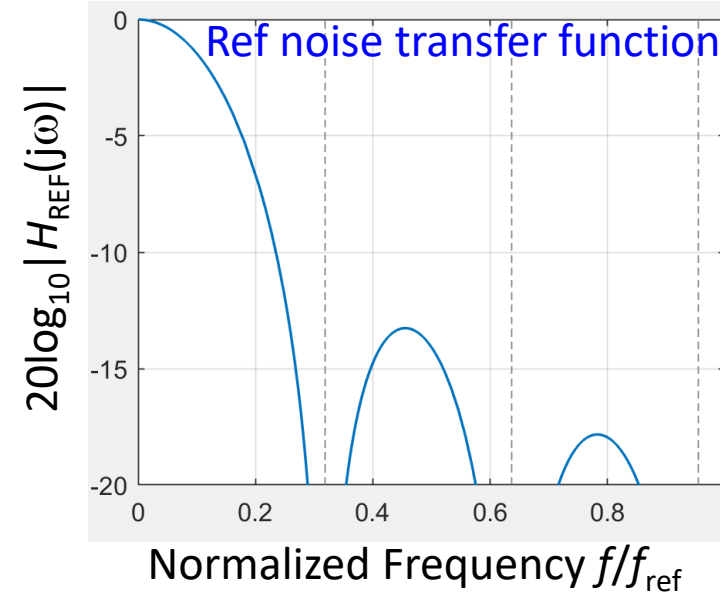
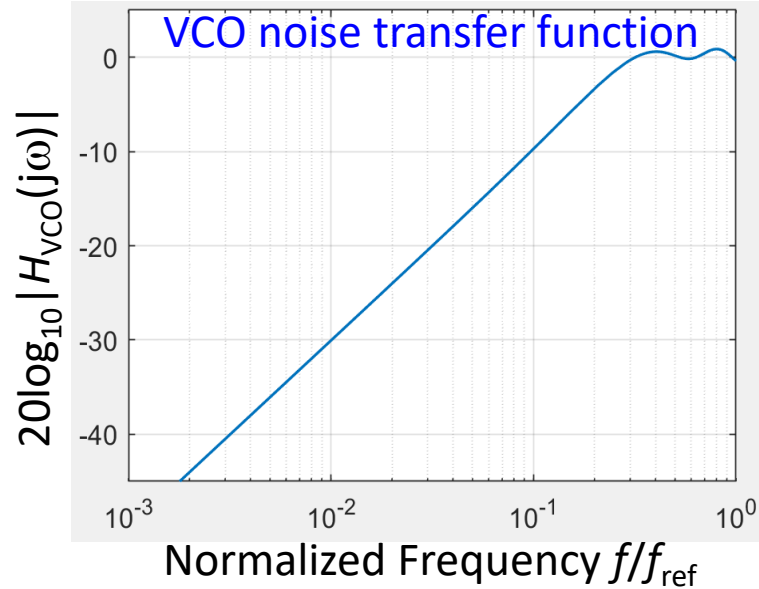
$$\theta_{out}(t) = \theta_{VCO}(t) + \sum_{n=-\infty}^{\infty} \varphi_{\Delta}[n] \cdot h_{hold}(t - nT_{ref}) \quad (1)$$

where $h_{hold}(t) = u(t) - u(t - T_{ref})$.

- The second term in (1) is a convolution of phase error $\varphi_{\Delta}[n]$ and hold function h_{hold} .



MDLL Phase Noise Transfer Functions



Perform Fourier transform on (1):

$$\theta_{out}(j\omega) = \theta_{VCO}(j\omega) + e^{-j\omega T_{ref}/2} \cdot \frac{\sin(\omega T_{ref}/2)}{\omega T_{ref}/2} \cdot \varphi_{\Delta}(z) \Big|_{z=e^{j\omega T_{ref}}}$$

$$\varphi_{\Delta}(j\omega) = N\theta_{ref}(j\omega) - \theta_{VCO}(j\omega)$$

$$\Rightarrow H_{VCO}(j\omega) = \frac{\theta_{out}(j\omega)}{\theta_{VCO}(j\omega)} = 1 - e^{-j\omega T_{ref}/2} \cdot \frac{\sin(\omega T_{ref}/2)}{\omega T_{ref}/2}$$

Ring oscillator phase noise transfer function

$$\Rightarrow H_{REF}(j\omega) = \frac{\theta_{out}(j\omega)}{\theta_{ref}(j\omega)} = N \cdot e^{-j\omega T_{ref}/2} \cdot \frac{\sin(\omega T_{ref}/2)}{\omega T_{ref}/2}$$

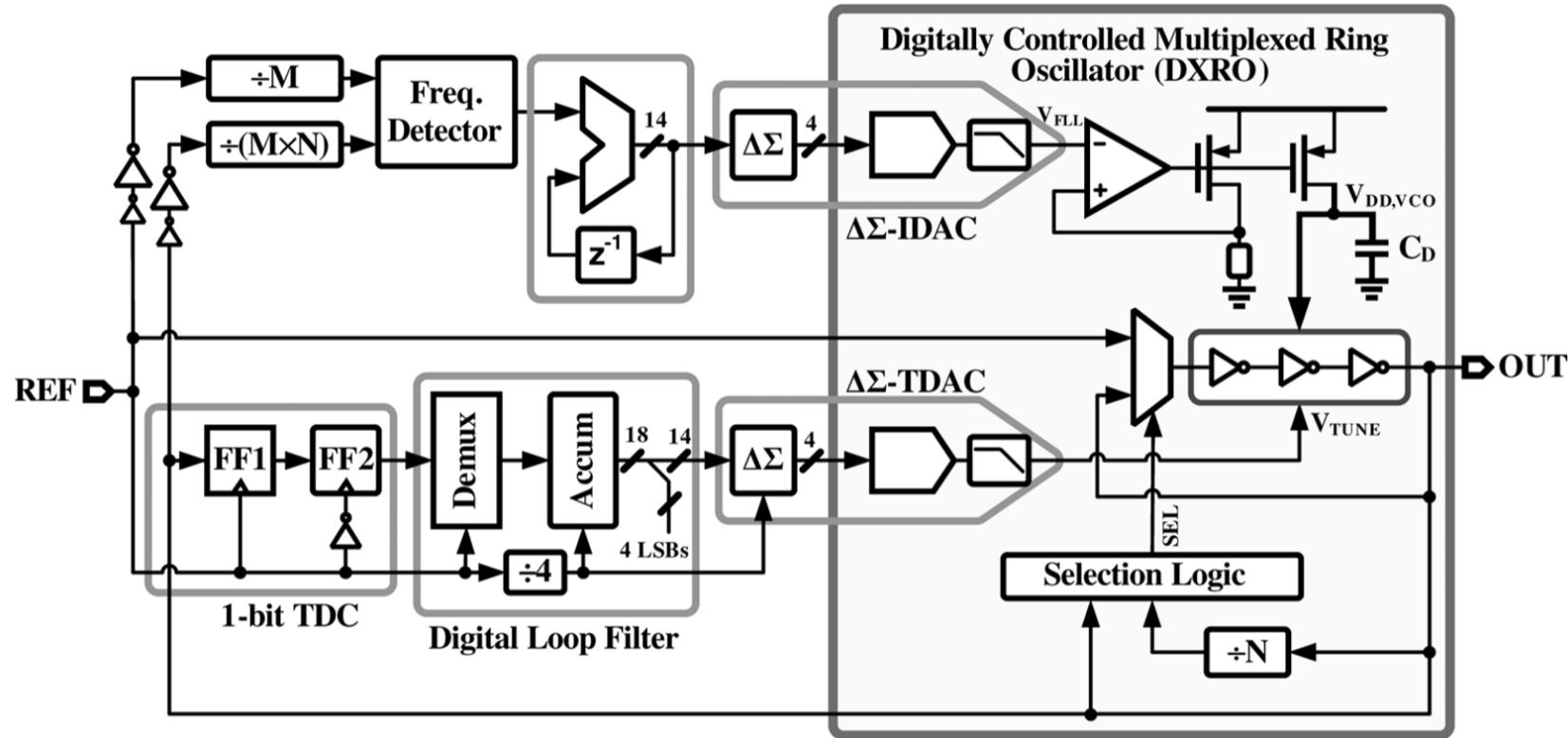
Reference phase noise transfer function

$$\theta_{out}(j\omega) = H_{VCO}(j\omega) \cdot \theta_{VCO}(j\omega) + H_{REF}(j\omega) \cdot \theta_{ref}(j\omega)$$

Ref[10]



Digital MDLL



- Digital frequency control loop
- Digital delay locked loop
 - TDC replaces PD
 - Digital loop filter replaces analog loop filter

Ref [11]



References

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